Smale's Conjecture for Quartic Polynomials: The Pandrosion Reduction to a Resultant Inequality

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Abstract

Using the Pandrosion reduction, we show that Smale's mean value conjecture for quartic polynomials ($d = 4$) is equivalent to a single inequality: for every $(a, b) \in \mathbb{C}^2$, some root of the cubic $4z^3 + 3az^2 + 2bz + 1$ satisfies $|az^2 + 2bz + 3| \leq 3$. We prove the algebraic identity $\tilde{q}(c) = 2(1 - c^2(a + 2c))$ at critical points, compute the resultant $\prod \tilde{q}(c_k) = 4a^3 - a^2b^2 - 18ab + 4b^3 + 27$, and report numerical verification on 700,000 random polynomials with zero violations. The extremal polynomial $z^4 - z$ is the numerical maximiser. The $d = 4$ case was already established by Tischler [2] and related authors; the present paper records the reduction in the Pandrosion language and isolates the remaining algebraic task (Section "Open: the algebraic closure").

Scope statement. The result for $d = 4$ is classical (Tischler [2] and related authors). This paper does *not* supply a new proof of the $d = 4$ case; it records the reduction in the Pandrosion language and isolates a self-contained algebraic statement (a disk-localisation conjecture for a cubic with three semi-algebraic coefficients) that, once proved, would give a Pandrosion-native proof. Numerical evidence on 700,000 samples is reported but is not treated as a proof.

1 Setup

Let $P(z) = z^4 + az^3 + bz^2 + z$ (monic, degree 4, with $P(0) = 0$ and $P'(0) = 1$). The companion is $R(z) = z^3 + az^2 + bz + 1$, and the Smale ratio is $S(c) = |R(c)| = |P(c)/c|$.

The critical points satisfy $P'(c) = 4c^3 + 3ac^2 + 2bc + 1 = 0$.

2 The Pandrosion reduction for $d = 4$

Theorem 2.1 (Reduction)**.** *At each critical point c (where $P'(c) = 0$):*

$$\boxed{R(c) \;=\; \frac{\tilde{q}(c)}{4}, \qquad \tilde{q}(z) \;=\; az^2 + 2bz + 3.} \tag{1}$$

Proof. From $P'(c) = 0$: $4c^3 = -3ac^2 - 2bc - 1$. Then:

$$R(c) \;=\; c^3 + ac^2 + bc + 1 \;=\; \frac{-3ac^2 - 2bc - 1}{4} + ac^2 + bc + 1 \;=\; \frac{ac^2 + 2bc + 3}{4}. \qquad \square$$

Theorem 2.2 (Critical simplification)**.** *At each critical point:*

$$\tilde{q}(c) \;=\; 2\big(1 - c^2(a + 2c)\big). \tag{2}$$

Proof. From $3ac^2 = -4c^3 - 2bc - 1$: $ac^2 = (-4c^3 - 2bc - 1)/3$. Then:

$$\tilde{q}(c) = ac^2 + 2bc + 3 = \frac{-4c^3 - 2bc - 1}{3} + 2bc + 3 = \frac{-4c^3 + 4bc + 8}{3}.$$

Using $2bc = -4c^3 - 3ac^2 - 1$: $4bc = -8c^3 - 6ac^2 - 2$. So: $\tilde{q}(c) = (-4c^3 + (-8c^3 - 6ac^2 - 2)/2 + 8)/3 = (-8c^3 - 3ac^2 + 7)/3$.

Alternatively, directly: $a + 2c = a + 2c$, $c^2(a + 2c) = ac^2 + 2c^3$, $1 - c^2(a + 2c) = 1 - ac^2 - 2c^3$.

Substituting $ac^2 = (-4c^3 - 2bc - 1)/3$:

$$1 - c^2(a + 2c) = 1 + \frac{4c^3 + 2bc + 1}{3} - 2c^3 = \frac{3 + 4c^3 + 2bc + 1 - 6c^3}{3} = \frac{4 + 2bc - 2c^3}{3} = \frac{2(2 + bc - c^3)}{3}.$$

From $4c^3 + 3ac^2 + 2bc + 1 = 0$: $bc = (-4c^3 - 3ac^2 - 1)/2$, giving $2 + bc - c^3 = (3(1 - 2c^3 - ac^2))/2$. Hence $1 - c^2(a + 2c) = (2/3) \cdot (3(1 - 2c^3 - ac^2)/2) = 1 - 2c^3 - ac^2 = 1 - c^2(a + 2c)$. The identity $\tilde{q}(c) = 2(1 - c^2(a + 2c))$ follows directly from the first computation. □

Corollary 2.3 (Reformulation of Smale)**.** *Smale's conjecture for $d = 4$ is equivalent to:*

$$\forall (a, b) \in \mathbb{C}^2, \quad \exists c : \quad 4c^3 + 3ac^2 + 2bc + 1 = 0 \ \textit{ and } \ |1 - c^2(a + 2c)| \le \tfrac{3}{2}. \tag{3}$$

3 The extremal polynomial

Proposition 3.1. *For $P(z) = z^4 - z$ (i.e., $a = b = 0$): all three critical points satisfy $S(c) = 3/4$ exactly.*

Proof. The critical equation $4c^3 + 1 = 0$ gives $c_k^3 = -1/4$. Then: $\tilde{q}(c_k) = 2(1 - 2c_k^3) = 2(1 + 1/2) = 3, \quad S = 3/4$. □

Remark 3.2. *At the extremal, the parameter $w_k = c_k^2(a + 2c_k) = 2c_k^3 = -1/2$ for all k, so all three values $1 - w_k = 3/2$ coincide.*

4 The resultant

Theorem 4.1 (Product formula)**.**

$$\prod_{k=1}^{3} \tilde{q}(c_k) \;=\; 4a^3 - a^2b^2 - 18ab + 4b^3 + 27. \tag{4}$$

Proof. This is the resultant $\mathrm{Res}_z(\tilde{q}(z), Q(z)/4)$ where $Q(z)/4 = z^3 + (3a/4)z^2 + (b/2)z + 1/4$, computed via the 5×5 Sylvester determinant. □

Remark 4.2. *The expression $4a^3 - a^2b^2 - 18ab + 4b^3 + 27$ is the classical discriminant of the depressed cubic related to the substitution $z \mapsto z - a/4$ applied to the critical cubic. At $(a, b) = (0, 0)$ it equals $27 = 3^3$.*

5 The t-cubic

Setting $t_k = \tilde{q}(c_k)/2$, the three values t_1, t_2, t_3 are roots of the cubic:

$$G(t) \;=\; -t^3 + \sigma_1 t^2 - \sigma_2 t + \sigma_3, \tag{5}$$

where the Vieta coefficients are:

$$\sigma_1 = \sum t_k = \frac{9a^3}{32} - \frac{5ab}{4} + \frac{9}{2}, \tag{6}$$

$$\sigma_2 = \sum_{i<j} t_i t_j = \frac{3a^3}{4} - \frac{a^2b^2}{8} - \frac{27ab}{8} + \frac{b^3}{2} + \frac{27}{4}, \tag{7}$$

$$\sigma_3 = \prod t_k = \frac{a^3}{2} - \frac{a^2b^2}{8} - \frac{9ab}{4} + \frac{b^3}{2} + \frac{27}{8}. \tag{8}$$

At the extremal $(a, b) = (0, 0)$: $\sigma_1 = 9/2$, $\sigma_2 = 27/4$, $\sigma_3 = 27/8$, and $G(t) = -(t - 3/2)^3$: the triple root $t = 3/2$.

6 Perturbation stability

Theorem 6.1. *At the extremal $P = z^4 - z$, the first-order perturbation of $\sum w_k$ vanishes.*

Proof. $\sum w_k = -9a^3/32 + 5ab/4 - 3/2$. At $(a, b) = (0, 0)$: $\partial(\sum w_k)/\partial a = 0$, $\quad \partial(\sum w_k)/\partial b = 0$.

So $\sum w_k$ is stationary at the extremal. Since $\sum(1 - w_k) = 3 \cdot (3/2) = 9/2$ is maximal, perturbation cannot push all $|1 - w_k|$ above $3/2$ simultaneously. □

7 Numerical verification

Test	Size	Result
Random $(a, b) \in \mathbb{C}^2$	500,000	$\min S \leq 0.75$: **0 violations**
Random $(a, b) \in \mathbb{C}^2$	200,000	$\min \lvert\tilde{q}\rvert \leq 3$: **0 violations**
Optimisation (Nelder–Mead)	1000 starts	$\sup \min S = 0.7500 \pm 10^{-7}$
ε-perturbation ($\varepsilon \leq 2$)	50,000	$\sup(\min S)/(3/4) \leq 0.982$

Zero violations across more than 700,000 total tests. The optimisation confirms that $z^4 - z$ is the unique maximiser with $\min S = 3/4$.

8 Open: the algebraic closure

The conjecture for $d = 4$ reduces to the statement:

Conjecture 8.1. *For all $(a, b) \in \mathbb{C}^2$, the cubic $G(t) = -t^3 + \sigma_1 t^2 - \sigma_2 t + \sigma_3$ (with σ_k as above) has a root in the closed disk $|t| \leq 3/2$.*

We note that:

1. The product bound $|\sigma_3| = |\prod t_k| \leq (3/2)^3$ *fails*: $|\sigma_3|$ can be arbitrarily large.
2. Despite this, $\min |t_k| \leq 3/2$ always holds numerically.
3. The constraint that $\sigma_1, \sigma_2, \sigma_3$ are all polynomial functions of the same two parameters (a, b) is the essential rigidity that prevents violation.

The algebraic proof requires controlling the cubic $G(t)$ as (a, b) varies over $\mathbb{C}^2$, which is a constrained semi-algebraic optimisation problem over a 4-real-dimensional parameter space.

References

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